CSC101: INTRODUCTION TO COMPUTING

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COURSE OUTLINE

- **Topics**
  - Digital Literacy
  - Computer Hardware
  - The Internet & Networks
  - Digital Safety and Security
  - Introduction to Python Programming
  - Digital Marketing

ASSESSMENT

- Continuous Assessment (40%)
  - Quizzes
  - Homework
  - Presentations
- Exams (60%)

LEARNING MANAGEMENT SYSTEM (E-LEARNING)



WHY DID YOU CHOOSE TO STUDY COMPUTER SCIENCE?

REASONS TO MAJOR IN COMPUTING (ACM)

1. Computing is part of everything we do!
2. Expertise in computing enables you to solve complex, challenging problems.
3. Computing enables you to make a positive difference in the world.
4. Computing offers many types of lucrative careers.

REASONS TO MAJOR IN COMPUTING (ACM)

5. Computing jobs are here to stay, regardless of where you are located.
6. Computing offers great opportunities for true creativity and innovativeness.
7. Computing is an essential part of well-rounded academic preparation.
8. Future opportunities in computing are without boundaries.

WHAT DO YOU WANT TO DO WITH A COMPUTER SCIENCE DEGREE?

CAREERS IN COMPUTING

- ICT Manager
- DB Administrator
- Systems Development Manager
- Systems Analyst
- Programmer
- Network Administrator
- Operations Manager
- Computer Operator
- Data Control Clerk
- Data Entry Operator
- File Librarian
- Data Security Analyst
- Computer Consultant
- EDP Auditor
- Webpage Designer
- Computer Engineer
- Software Engineer
- Computer Technician
- Software Tester
- Software Trainer
- Multimedia Artist and Animators
- Computer Sales Rep.

WHAT IS THE DIFFERENCE BETWEEN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

CS VS IT

- Computer science refers to designing and building computers and computer programs.

- Information technology refers to maintaining and troubleshooting those computers and their networks, systems, and databases to ensure they run smoothly.

CS VS IT

- Information technology) careers are more about installing, maintaining, and improving computer systems, operating networks, and databases.

- Computer science (CS) focuses on using mathematics to program systems to run more efficiently, including in design and development.

CS VS IT

- Computer science professionals use mathematics to develop and improve computer programs.
- IT professionals install, implement, and maintain these programs and apply them to businesses.
- Computer science professionals use skills in mathematics and theories to make information technology possible
- IT professionals use skills to communicate with businesses to meet their needs through technology.

COMPUTER SCIENCE VS. IT CAREERS

- Jobs grounded in computer science will broadly mean working with programming languages or data to develop or improve products.

- Jobs in IT will focus more heavily on the operation of computers, their networks, and systems, so that others in an organization can do their work smoothly.

Computer Science classes

- Theory of computation
- Software security
- Computer graphics
- Algorithms and data structures
- Calculus
- Linear algebra
- Software engineering
- Web development
- Cryptography
- Programming

Information Technology classes

- Information Systems
- Network Engineering
- Operating Systems
- Database Fundamentals
- Information Security Fundamentals
- System Analysis
- Business Technology
- Hardware and Software Troubleshooting
- Network Administration
- IT Ethics

STRUCTURE OF CS PROGRAM (LEVEL 100)

CSC 101: Introduction to Computing
CSC 102 Programming Techniques
CSC 107: Software Suite
PHY 101: General Physics I (Theory)
PHY 103: General Physics I (Practical)
MAT101: Algebra and Trigonometry
MAT 102: Analytic Geometry and Calculus
PHY 102: General Physics II (Theory)
PHY 104: General Physics II (Practical)
STA 102: Introduction to Statistics

STRUCTURE OF CS PROGRAM (LEVEL 200)

CSC 203: Programming and Problem Analysis I
CSC 211: Computer Architecture
CSC 213: Database Management I
MAT 201 Introduction to Abstract Algebra
MAT 203: Further Calculus
CSC 212: Algorithms
CSC 214: Operating Systems
CSC 216: Database Management II
MAT 202: Vector Algebra and Differential Equations
STA 202: Further Statistics

STRUCTURE OF CS PROGRAM (LEVEL 300)

CSC 311: Computer Security
CSC 315: Discrete Mathematics
CSC 317: Software Engineering
CSC 319: Data Structures:
MAT 301: Advanced Calculus I
CSC 302: Programming and Problem Analysis II (Java)
CSC 312: Human Computer Interface
CSC 314: Network Computing I
CSC 316: Web Technologies
CSC 399: Research Methods

STRUCTURE OF CS PROGRAM (LEVEL 300)

CSC 401: Introduction to Compilers
CSC 405: Advanced Operating Systems
CSC 411: Introduction to Artificial Intelligence
CSC 413: Computer Graphics
CSC 415: Network Computing II
CSC 402: Numerical Analysis
CSC 406: Systems Administration
CSC412: Cryptography and Network Security
CSC 408: Operations Research
CSC 499: Project Work

Computer science jobs	Salary (average base pay in US, Glassdoor)	IT jobs	Salary (average base pay in US, Glassdoor)
Web developer	$82,690	Computer technician	$45,809
Robotics engineer	$101,108	Helpdesk technician	$49,743
Full stack developer	$87,080	Cybersecurity specialist	$89,115
Software engineer	$105,772	System administrator	$78,645
Artificial intelligence engineer	$125,124	Database administrator	$84,082
Back end engineer	$126,948	Network administrator	$71,311
Data scientist	$124,223	Cloud engineer	$151,884

Computer science skills	IT skills
Programming languages like Python, C++, HTML, CSS, Javascript	Programming languages, especially scripting languages like Python and Powershell
Statistics, algebra, and/or calculus	Troubleshooting methods
Familiarity with code sharing platforms like GitHub	Network configuration
Understanding of the software development lifecycle	Security infrastructure like setting up firewalls and routers
Artificial intelligence methods like machine learning and deep learning, and tools like Apache Spark and Hadoop	Ethical hacking or penetration testing
Data analysis and database tools like SQL, MongoDB, PostgreSQL, Excel	Data administration skills like SQL

COMPUTER SCIENCE SPECIALIZATION

- **Artificial Intelligence**
  - **Skills:** Mathematics and analysis, algorithms, predictive modeling
  - **Common Roles:** Artificial Intelligence Architect, Artificial Intelligence Researcher, Machine Learning Engineer

- **Computer-Human Interface**
  - **Skills:** Communication, interpersonal skills, attention to visual detail, mapping how people use software and systems
  - **Common Roles:** User Experience (UX) Designer, User Experience Researcher, Interaction Designer

COMPUTER SCIENCE SPECIALIZATION

- **Computer Graphics**
  - **Skills:** Attention to visual and artistic detail, collaboration, creativity
  - **Common Roles:** Mobile Application Developer, iOS Developer, Android Developer
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- **Information Security**
  - **Skills:** Communication, threat/vulnerability management, knowledge of security compliance rules and regulations
  - **Common Roles:** Security Engineer, Network Security Analyst, System Security Analyst

COMPUTER SCIENCE SPECIALIZATION

- **Programming Languages**
  - **Skills:** Coding and scripting in multiple languages, collaboration
  - **Common Roles:** Full-Stack Web Developer, Front-End Developer, Game Developer, Software Programmer

- **Software Engineering**
  - **kills:** Coding and scripting, communication, collaboration
  - **Common Roles:** Software Development Engineer, Software Engineer

COMPUTER SCIENCE SPECIALIZATION

- **Game Design**
  - **kills:** Attention to visual detail, collaboration, coding, and scripting
  - **Common Roles:** Video Game Designer, Video Game Developer, Software Engineer

- **Data Science**
  - **Skills:** Mathematics and analytics, attention to detail, predictive modeling
  - **Common Roles:** Data Scientist, Data Analyst, Business Intelligence Analyst, Machine Learning Engineer, Information Scientist, Database Administrator

COMPUTER SCIENCE SPECIALIZATION

- **Networks**
  - **Skills:** Diagnose and troubleshoot network issues, design network architecture
  - **Common Roles:** Network Administrator, Network Analyst, Network Architect

- **Theory**
  - **Skills:** Mathematics and analytics, predictive modeling and probability, big-picture planning
  - **Common roles:** Algorithm Scientist, Machine Learning Engineer

ASSIGNMENT

- What is Artificial intelligence?
- List 10 applications of Artificial intelligence?
- Discuss roles, benefits and drawbacks of artificial intelligence?
- Deadline: January 31, 2023

- Do not exceed 2 pages